

CONDUCTOR & SYMPHONY NODES

Conductor and Symphony Nodes operate together as an orchestrated network for executing and managing tasks across Huey's distributed system. The Conductor Node acts as the control authority, assigning and balancing workloads, while Symphony Nodes perform the actual processing and operations.

Technical Specifications:

- Conductor Node – High-availability node responsible for issuing instructions and managing inter-node communication latency within predefined SLAs.
- Symphony Nodes – Distributed compute agents optimized for parallel processing, redundancy, and local decision-making within assigned boundaries.

Governance Integration:

- All task assignments and node role changes are recorded in the Unified Audit Log (UAL) with actor, timestamp, and scope of changes.
- Governance approval is required for reassignment of conductor duties or cross-domain tasking.

Operational Advantages:

- Clear separation of coordination and execution roles.
- Improved resilience via distributed execution and redundant command channels.
- Scalable system architecture that can dynamically adjust the ratio of conductors to symphonies.

KPIs:

- Coordination latency (target ≤ 200 ms).
- Task throughput per Symphony Node.
- Recovery time after conductor node failure.

Testing Objectives:

- Stress test under high task volume to measure coordination efficiency.
- Simulate conductor failure and observe symphony autonomy until recovery.
- Validate governance oversight during large-scale task redistribution.

Cross-References:

- See 'Hierarchy Models' for layered control framework.
- See 'Orchestration' for workload deployment processes.

[DIAGRAM PLACEHOLDER: 'Node Hierarchy and Orchestration Command Flow' — showing conductor issuing commands, symphony nodes executing workloads, and governance feedback loops]